

# Robust Design and Experimental Demonstration of RSS-Based Visible Light Positioning Under Limited Receiving FOV

Nuo Huang<sup>✉</sup>, Feng Chen<sup>✉</sup>, Changyuan Liu, Yan-Yu Zhang, Yi-Jun Zhu<sup>✉</sup>,  
and Chen Gong<sup>✉</sup>, *Senior Member, IEEE*

**Abstract**—In this paper, we propose a robust received signal strength (RSS)-based indoor visible light positioning (VLP) scheme under limited receiving field-of-view (FOV). The position estimation in the time domain is complicated, and the time-frequency transform is first performed on the received time-domain signal to establish an equivalent frequency-domain RSS model, which simplifies the signal representation and problem solving. To realize robust VLP under limited receiving FOV, we consider the joint estimation of the receiver coordinate, light emitting diode (LED) validity indicator vector, signal phase vector and receiving coefficient based on the frequency-domain RSS vector. The original estimation problem is equivalently transformed into the optimization problem that maximizes the angle between the valid RSS modulus vector and the valid RSS feature vector. Experimental and simulation results verify the effectiveness of the proposed scheme under limited receiving FOV. Specifically, experimental results show that the proposed VLP scheme achieves a stable positioning accuracy below 5 cm under limited receiving FOV in a 60 cm × 60 cm positioning region with transceiver height 150 cm.

**Index Terms**—Visible light positioning, indoor positioning, received signal strength, limited field-of-view.

## I. INTRODUCTION

WITH the rapid development of location-based applications, high-accuracy positioning techniques have drawn great interest from both academia and industry [1], [2], [3], [4]. Visible light positioning (VLP) based on light emitting diode (LED) is considered as a promising indoor positioning technique due to the distinct advantages of high positioning accuracy and low implementation cost [5], [6], [7]. Based on the difference of receiver devices, the VLP technologies

can be classified into two categories, namely imaging VLP and non-imaging VLP. Imaging VLP typically estimates the terminal position by processing the images of LEDs captured by the cameras [8], [9], [10], [11]. Imaging-based VLP can achieve higher positioning accuracy, but brings higher implementation complexity and larger privacy risks than non-imaging VLP [12]. Non-imaging VLP typically utilizes the photodiode (PD) to extract optical features such as time of arrival [13], angle of arrival [14], and received signal strength (RSS) [15]. Compared with the other non-imaging VLP technologies, RSS-based VLP is the most popular candidate in indoor positioning systems since RSS information can be easily acquired with no requirement of auxiliary devices [16].

There have been extensive studies on indoor RSS-based VLP. Works [17], [18], [19] estimated the terminal position by exploiting the received optical power, i.e., direct current (DC) signal strength. Nevertheless, the positioning accuracy of a DC-based VLP scheme is susceptible to the ambient light, which is less practical in real scenarios. Meanwhile, there is a lack of experimental verification for these positioning schemes based on DC signal strength. Also note that there can be a trade-off to select the FOV of PD in VLP system, especially when DC signal strength is employed for positioning. For instance, the receiver with a smaller FOV could not receive enough light sources to provide a satisfactory positioning accuracy, while the receiver with a larger FOV could introduce a greater amount of ambient light [20], [21], [22].

Alternating current (AC) signal strength can be employed for VLP to alleviate the effect of ambient light, as the ambient light is typically a near-DC component and can be well separated with the AC signal component for VLP. Work [23] proposed an AC-based VLP scheme, which estimates the receiver position from the transmission distances and can experimentally achieve a 2-dimensional (2D) positioning accuracy about 2 cm in a 20 × 20 cm<sup>2</sup> square region. Nonlinear optimization method was adopted in [24] to improve the positioning accuracy, which experimentally achieves a 2D positioning error about 3 cm in a 5 × 5 m<sup>2</sup> square region adopting high-power LEDs. Signal strength enhancement and convolutional neural network were employed in [25] to alleviate the LED deficient regions in VLP, which experimentally achieves a 2D positioning error about 5 cm in a 1.5 × 2 m<sup>2</sup> square region. The VLP methods in [23], [24], [25] are highly dependent on the values of transmission parameters,

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Nuo Huang, Feng Chen, Changyuan Liu, and Chen Gong are with the CAS Key Laboratory of Wireless-Optical Communications, School of Information Science and Technology, University of Science and Technology of China, Hefei 230027, China (e-mail: huangnuo@ustc.edu.cn; cf10015@mail.ustc.edu.cn; liuchangyuan@mail.ustc.edu.cn; cgong821@ustc.edu.cn).

Yan-Yu Zhang and Yi-Jun Zhu are with the School of Information System Engineering, Information Engineering University, Zhengzhou 450000, China (e-mail: xindazhang@126.com; yijunzhu@zzu.edu.cn).

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and imperfect knowledge of transmission parameters can greatly increase the positioning error. By jointly estimating the receiver position and receiver coefficient, work [26] proposed a robust VLP scheme which experimentally achieves a stable 2D positioning accuracy about 4 cm in a  $60 \times 60$  cm<sup>2</sup> square region under different transmission coefficients. Work [27] achieved a positioning accuracy of 1 cm using 25 LEDs and extended Kalman filter, which requires exact knowledge of power distribution in the room and is only demonstrated via simulations.

Existing studies typically assume that the receiving field-of-view (FOV) of PD is sufficiently large, and adopt all LEDs for positioning. However, the FOV of PD can be limited by the package or concentrator [28], [29], [30]. When the incidence angle is larger than the receiving FOV, the transmitted signal from the corresponding LED cannot be received by the PD and the PD output is merely the noise (such LED is termed as “invalid LED” in this work, while the LED whose transmitted signal can be received effectively by the PD is termed as “valid LED”). In this case, if the invalid LEDs are considered as valid LEDs (i.e., the noise is mistakenly adopted as useful signal), the positioning accuracy can be degraded due to the ineffective RSS information, which makes the current RSS-based VLP schemes less practical in real applications.

To address the above-mentioned issue, this work proposes a robust RSS-based VLP scheme under limited receiving FOV. For the convenience of signal representation and problem solving, the time-frequency transform is firstly performed on the received time-domain signal to obtain an equivalent frequency-domain RSS model. For robust VLP using the frequency-domain RSS vector, we formulate the joint estimation problem with respect to the receiver coordinate, light emitting diode (LED) validity indicator vector, signal phase vector and receiving coefficient. The original estimation problem is equivalently transformed into the problem that maximizes the angle between the valid RSS modulus vector and the valid RSS feature vector. The proposed scheme is robust under limited receiving FOV due to the LED validity determination, and only requires the power ratios and attenuation exponents of the transmitters. Experimental and simulation results demonstrate the superiority of the proposed scheme over the benchmark scheme adopting fixed number of LEDs under limited FOV. Specifically, experimental results show that the proposed VLP scheme achieves a stable positioning accuracy below 5 cm under a limited receiving FOV in a  $60 \text{ cm} \times 60 \text{ cm}$  positioning region with transceiver height 150 cm. Also note that the positioning accuracy of our proposed scheme can be further improved by increasing the transmission power or the number of LEDs. To the best of our knowledge, this is the first experimental demonstration of robust RSS-based VLP under limited receiving FOV.

The remainder of this paper is organized as follows. The considered VLP system and signal model are outlined in Section II. Section III elaborates the proposed VLP scheme. Experimental and simulation results are provided in Sections IV and V. Finally, Section VI concludes this paper.

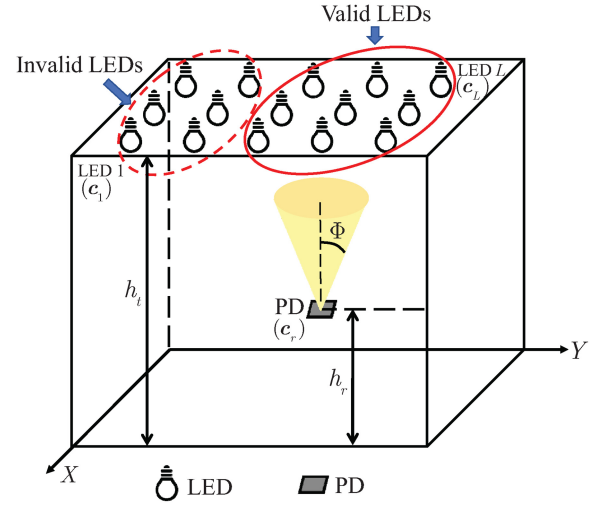


Fig. 1. Illustration of the considered indoor VLP scenario.

## II. SYSTEM DESCRIPTIONS

Figure 1 shows the considered indoor VLP system with  $L$  LEDs at the transmitter and one PD at the receiver. Assume that the LEDs and PD are all parallel to the horizontal plane, which is reasonable in common positioning applications [31], [32], [33], [34]. All LEDs are deployed on the ceiling at the same height  $h_t$ , and the 3-dimensional (3D) coordinate of LED  $l$  is known as  $\mathbf{c}_l = [x_l, y_l, h_t]^T \in \mathbb{R}^3$ ,  $l = 1, \dots, L$ . The PD is installed on the receiver terminal with known height  $h_r$ , and its 3D coordinate is denoted by  $\mathbf{c}_r = [x_r, y_r, h_r]^T \in \mathbb{R}^3$ , where the horizontal coordinate  $(x_r, y_r)$  needs to be estimated. The distance between LED  $l$  and the PD is calculated as  $d_{r,l} = \|\mathbf{c}_l - \mathbf{c}_r\| = \sqrt{(x_l - x_r)^2 + (y_l - y_r)^2 + h^2}$  with  $h = h_t - h_r$ , where  $\|\cdot\|$  denotes the  $l_2$ -norm of a vector. The considered VLP system can model the scenario where the receiver is fixed on a mobile platform (e.g., a robot).

To eliminate the mutual interference, each LED transmits its own beacon signal comprising DC and AC components to the PD in different time slots. For alleviating the effect of ambient light, this work realizes VLP using the AC components which are chosen to be the sinusoidal signals with frequency  $f$ . In the  $l$ -th time slot, the PD receives the optical signal from LED  $l$  and converts it into an electrical signal. After filtering out the DC component via a DC-removal circuit, the output time-domain AC signal of PD can be written as

$$p_{r,l}(t) = RH_{r,l}p_l \cdot \cos(2\pi f \cdot t + \phi_{r,l}) + z_{r,l}(t), \quad (1)$$

where  $p_l$  denotes the emitted AC signal strength from LED  $l$ ;  $H_{r,l}$  denotes the optical channel gain from LED  $l$  to the PD;  $\phi_{r,l}$  is the signal phase;  $R$  is the detection gain of PD; and  $z_{r,l}(t)$  denotes the additive Gaussian noise with zero mean and variance  $\sigma^2$ . From the Lambertian model, the optical channel gain  $H_{r,l}$  is expressed as [35]

$$H_{r,l} = \frac{A_r(m_l + 1)}{2\pi\|\mathbf{c}_l - \mathbf{c}_r\|_2^2} \cos^{m_l}(\theta_l) \cos(\varphi_{r,l}) \text{rect}\left(\frac{\varphi_{r,l}}{\Phi}\right), \quad (2)$$

where  $A_r$  is the photosensitive area of PD;  $m_l$  denotes the Lambertian order of LED  $l$ ;  $\theta_l$  is the irradiance angle of LED  $l$ ;  $\varphi_{r,l}$  denotes the incidence angle;  $\Phi$  is the FOV of PD;

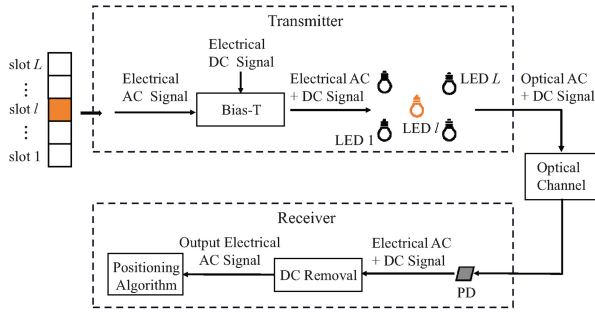


Fig. 2. Diagram of the considered VLP system.

and  $\text{rect}(\cdot)$  is the rectangular function. Since all LEDs and PD are parallel to the horizontal plane, we have  $\cos(\theta_l) = \cos(\varphi_{r,l}) = \frac{h}{d_{r,l}}$ .

Existing studies typically assume that the receiving FOV of PD is sufficiently large. However, the FOV of PD can be limited by the package or concentrator [28], [29], [30]. According to [36], [37], [38], the FOV of PD may fall in the range of  $[10^\circ, 80^\circ]$ . From Eq. (2), the PD cannot receive the transmitted signals and its output is merely the noise when the incidence angle is larger than the receiving FOV. In this work, we term the LED whose transmitted signal cannot be received by the PD as “invalid LED”, and the LED whose transmitted signal can be received effectively by the PD as “valid LED”. If the invalid LEDs are considered as valid LEDs (i.e., the noise is mistakenly adopted as useful signal), the positioning accuracy can be degraded due to the ineffective RSS information, which motivates the research of this work. Also note that the positioning accuracy improves with the number of valid LEDs adopted for VLP, since signals from additional LEDs can provide more information about the receiver position. We use binary variable  $t_{r,l}$  to indicate the validity of LED  $l$ , i.e.,  $t_{r,l} = 1$  if LED  $l$  is valid and  $t_{r,l} = 0$  otherwise. Signal  $p_{r,l}(t)$  is sampled with frequency  $f_s$  at the receiver. For notational simplicity, we define the attenuation exponent of LED  $l$  as  $M_l = m_l + 3$ , the LED validity indicator vector as  $\mathbf{t}_r = [t_{r,1}, \dots, t_{r,L}]^T$ , the signal phase vector as  $\boldsymbol{\phi}_r = [\phi_{r,1}, \dots, \phi_{r,L}]^T$ , the transmitter characteristic scalar as  $g_l = (m_l + 1)p_l$ , the receiver characteristic scalar as  $g_r = RA_r/2\pi h^2$ , and the transceiver coefficient as  $U_{r,l} = g_r g_l (h/d_{r,l})^{M_l}$ . Combining (1) and (2), the discrete-time samples can be written as

$$\begin{aligned} p_{r,l}[n] &= g_r g_l \left( \frac{h}{d_{r,l}} \right)^{M_l} \cos \left( 2\pi \frac{f}{f_s} \cdot n + \phi_{r,l} \right) \cdot t_{r,l} + z_{r,l}[n] \\ &= U_{r,l} \cos \left( 2\pi \frac{f}{f_s} \cdot n + \phi_{r,l} \right) \cdot t_{r,l} + z_{r,l}[n], \\ n &= 1, \dots, N, \end{aligned} \quad (3)$$

where  $N$  is the number of signal samples.

In the considered VLP system, the receiver position  $(x_r, y_r)$  is estimated from the collected discrete-time samples  $\{p_{r,l}[n]\}_{l=1, n=1}^{L, N}$ . The overall diagram of the VLP system is shown in Fig. 2, where the detailed positioning algorithm will be given in Section III-B. Note that transmitter parameters  $\{g_l\}_{l=1}^L$  and  $\{M_l\}_{l=1}^L$  can be considered as the prior knowledge, since the LEDs are generally fixed in real VLP

scenarios. Besides, the values of  $\{g_l\}_{l=1}^L$  may be different from each other even if all LEDs are driven by the same DC and AC signals. Considering the variations of receiver device and receiver position in a VLP system, it is rational to assume that the receiving characteristic scalar  $g_r$ , LED validity indicator vector  $\mathbf{t}_r$  and signal phase vector  $\boldsymbol{\phi}_r$  are unknown. Traditional VLP schemes typically assume  $t_{r,l} = 1$  for all  $l$  and neglect the effect of limited receiving FOV, which lead to a performance degradation and are less practical in real scenarios. To address this issue, this work proposes a robust VLP scheme under limited receiving FOV, where only the knowledge on the values of  $\{M_l\}_{l=1}^L$  and ratios of  $\{g_l\}_{l=1}^L$  are available. The values of  $\{M_l\}_{l=1}^L$  and ratios of  $\{g_l\}_{l=1}^L$  can be estimated using the method presented in [26].

*Remark 1:* In the considered VLP scenario, the positioning algorithm is processed at the receiver side (i.e., client terminal). Generally, the positioning accuracy of a VLP scheme can be further improved by increasing the transmission power, the number of LEDs or the number of PDs. To reduce the implementation cost and computational complexity at the client terminal in real application, only a single PD is utilized for VLP in this work, similarly as in many existing works (e.g., [15], [19], [20], [23]). Meanwhile, we focus on the solution to the performance degradation due to the ineffective RSS information when incorrectly using invalid LED as valid LED.

*Remark 2:* Compared to the case with non-tilted LEDs, the positioning accuracy in the center region can be improved by using tilted LEDs, but that in the edge region is degraded due to the decrease of signal power [39]. Following many existing works, we assume that the LEDs and PD are all parallel to the horizontal plane, which enjoys good compatibility with existing illumination infrastructure. The impact of small LED tilting angles can be alleviated by the calibration of LED transmission parameters, which is performed in our experiments and verified to achieve a stable positioning accuracy (cf. Section IV).

### III. PROPOSED VLP SCHEME

In this section, we propose a VLP scheme based on the time-frequency transform and LED validity determination.

#### A. Time-Frequency Transform of Received Signals

Note that the position estimation problem based on the time-domain samples  $\{p_{r,l}[n]\}_{l=1, n=1}^{L, N}$  is complicated due to the large dimension and coupling relationship of  $\{p_{r,l}[n]\}_{l=1, n=1}^{L, N}$ . For simplification of signal representation and problem solving, we first perform  $N$ -point discrete Fourier transform (DFT) on the time-domain signal in (3), yielding the frequency-domain signal [40]:

$$P_{r,l}[k] = \frac{1}{N} \sum_{n=0}^{N-1} p_{r,l}[n] \cdot e^{-j \frac{2\pi k}{N} n}, \quad k = 0, \dots, N-1. \quad (4)$$

Define the useful time-domain signal component in  $p_{r,l}[n]$  as

$$x_{r,l}[n] = U_{r,l} \cos \left( 2\pi \frac{f}{f_s} \cdot n + \phi_{r,l} \right). \quad (5)$$

The DFT of  $x_{r,l}[n]$  is given by

$$\begin{aligned} X_{r,l}[k] &= \frac{1}{N} \sum_{n=0}^{N-1} x_{r,l}[n] \cdot e^{-j\frac{2\pi k}{N}n} \\ &= \frac{1}{N} \sum_{n=0}^{N-1} U_{r,l} \cos\left(2\pi \frac{f}{f_s} \cdot n + \phi_{r,l}\right) \cdot e^{-j\frac{2\pi k}{N}n}, \\ k &= 0, \dots, N-1. \end{aligned} \quad (6)$$

Denoting the frequency index  $a = \frac{Nf}{f_s}$  with  $\frac{f}{f_s} > 2$ , we have

$$X_{r,l}[k] = \begin{cases} \frac{U_{r,l}}{2} e^{j\phi_{r,l}}, & k = a \\ 0, & k \neq a \end{cases}. \quad (7)$$

According to (7), we only need to consider  $X_{r,l}[a]$  for VLP, which simplifies the signal representation and problem solving.

The frequency-domain noise component is given by

$$Z_{r,l}[k] = \frac{1}{N} \sum_{n=0}^{N-1} z_{r,l}[n] \cdot e^{-j\frac{2\pi k}{N}n}, \quad k = 0, \dots, N-1. \quad (8)$$

Denote the frequency-domain noise vector by  $\mathbf{Z}_{r,l} = [Z_{r,l}[0], \dots, Z_{r,l}[N-1]]^T$ . The covariance matrix of  $\mathbf{Z}_{r,l}$  is  $\mathbb{E}[\mathbf{Z}_{r,l} \mathbf{Z}_{r,l}^H] = \frac{\sigma_z^2}{N} \mathbf{I}_{N \times N}$ .

Using (4)–(7), the time-domain RSS model in (3) can be equivalently transformed into the following frequency-domain RSS model:

$$\begin{aligned} P_{r,l}[a] &= X_{r,l}[a] \cdot t_{r,l} + Z_{r,l}[a] \\ &= \frac{U_{r,l}}{2} e^{j\phi_{r,l}} \cdot t_{r,l} + Z_{r,l}[a] \\ &= \frac{1}{2} g_r g_l \left(\frac{h}{d_{r,l}}\right)^{M_l} e^{j\phi_{r,l}} \cdot t_{r,l} + Z_{r,l}[a] \\ &= g_r s_l \cdot T_{r,l} + Z_{r,l}[a], \end{aligned} \quad (9)$$

where  $s_l = \frac{g_l}{2} (h/d_{r,l})^{M_l}$  and  $T_{r,l} = e^{j\phi_{r,l}} \cdot t_{r,l}$ . For simplicity, the frequency index  $[a]$  in (9) is omitted in the following derivation. Comparing (3) and (9), the  $N$ -length time-domain vector  $[p_{r,l}[1], p_{r,l}[2], \dots, p_{r,l}[N]]^T$  is equivalently represented by the frequency-domain scalar  $P_{r,l}$ , simplifying the subsequent processing. Under the assumption of complex Gaussian noise in (9), the conditional probability density function (PDF) of  $P_{r,l}$  given  $x_r, y_r, g_r$  and  $T_{r,l}$  is

$$\begin{aligned} f(P_{r,l}|x_r, y_r, g_r, T_{r,l}) &\propto \\ \exp\left\{-\frac{(P_{r,l} - g_r s_l T_{r,l})^H (P_{r,l} - g_r s_l T_{r,l})}{2\sigma^2/N}\right\}. \end{aligned} \quad (10)$$

## B. Positioning Based on LED Validity Determination

Denote the frequency-domain RSS vector by  $\mathbf{P}_r = [P_{r,1}, \dots, P_{r,L}]^T$ , the RSS characteristic vector by  $\mathbf{s} = [s_1, \dots, s_L]^T$ , and the LED indicator characteristic vector by  $\mathbf{T}_r = [T_{r,1}, \dots, T_{r,L}]^T$ . We consider joint maximum-likelihood (ML) estimation of the receiver horizontal coordinate  $(x_r, y_r)$ , receiver coefficient  $g_r$ , signal phase

vector  $\boldsymbol{\phi}_r$  and the LED validity indicator vector  $\mathbf{t}_r$ . With (9), the joint ML estimation problem can be formulated as

$$\begin{aligned} &(\hat{x}_r, \hat{y}_r, \hat{g}_r, \hat{\boldsymbol{\phi}}_r, \hat{\mathbf{t}}_r) \\ &= \arg \max_{x_r, y_r, g_r, \boldsymbol{\phi}_r, \mathbf{t}_r} \sum_{l=1}^L \ln f(P_{r,l}|x_r, y_r, g_r, T_{r,l}) \\ &= \arg \min_{x_r, y_r, g_r, \boldsymbol{\phi}_r, \mathbf{t}_r} \|\mathbf{P}_r - g_r \mathbf{s} \odot \mathbf{T}_r\|^2, \end{aligned} \quad (11)$$

where  $\odot$  denotes Hadamard product. There are  $2L + 3$  optimization variables in problem (11), and it can be simplified by reducing the number of jointly estimated unknowns.

Given  $\hat{x}_r, \hat{y}_r, \hat{g}_r$  and  $\hat{\mathbf{t}}_r$ , problem (11) becomes

$$\hat{\boldsymbol{\phi}}_r = \arg \min_{\boldsymbol{\phi}_r} \|\mathbf{P}_r - g_r \mathbf{s} \odot \mathbf{T}_r\|^2. \quad (12)$$

Problem (12) can be decoupled into the following  $L$  independent subproblems:

$$\hat{\phi}_{r,l} = \arg \min_{\phi_{r,l}} |P_{r,l} - g_r s_l T_{r,l}|^2, \quad l = 1, \dots, L. \quad (13)$$

The solution to problem (13) is readily obtained as

$$\begin{aligned} \hat{\phi}_{r,l} &= \arctan \frac{\text{Im}\left(\frac{P_{r,l}}{g_r s_l T_{r,l}}\right)}{\text{Re}\left(\frac{P_{r,l}}{g_r s_l T_{r,l}}\right)} = \arctan \frac{\text{Im}(P_{r,l})}{\text{Re}(P_{r,l})} \\ &= \angle P_{r,l}. \end{aligned} \quad (14)$$

Using (14) and denoting the modulus of  $P_{r,l}$  by  $\tilde{P}_{r,l} = |P_{r,l}|$ , problem (11) is equivalent to

$$\begin{aligned} (\hat{x}_r, \hat{y}_r, \hat{g}_r, \hat{\mathbf{t}}_r) &= \arg \min_{x_r, y_r, g_r, \mathbf{t}_r} \sum_{l=1}^L \left(\tilde{P}_{r,l} - g_r s_l t_{r,l}\right)^2 \\ &= \arg \min_{x_r, y_r, g_r, \mathbf{t}_r} \|\tilde{\mathbf{P}}_r - g_r \mathbf{s} \odot \mathbf{t}_r\|^2, \end{aligned} \quad (15)$$

where  $\tilde{\mathbf{P}}_r = [\tilde{P}_{r,1}, \dots, \tilde{P}_{r,L}]^T$  denotes the frequency-domain RSS modulus vector.

Denote the number of valid LEDs by  $L_v \in \{1, \dots, L\}$ . With  $t_{r,l} = \text{rect}(\frac{\phi_{r,l}}{\Phi})$ , it is obvious that the valid LEDs are those corresponding to the  $L_v$  smallest incidence angles. Based on this fact, we sort the incidence angles in the non-decreasing order such that

$$\varphi_{I(1)} \leq \dots \leq \varphi_{I(L_v)} \leq \dots \leq \varphi_{I(L)}. \quad (16)$$

Denote the frequency-domain valid RSS modulus vector by  $\tilde{\mathbf{P}}_{r,I} = [\tilde{P}_{r,I(1)}, \dots, \tilde{P}_{r,I(L_v)}]^T$  and the valid RSS characteristic vector by  $\mathbf{s}_I = [s_{I(1)}, \dots, s_{I(L_v)}]^T$ . Through the above analysis, problem (15) can be simplified by replacing  $\mathbf{t}_r$  with  $L_v$  as

$$\begin{aligned} &(\hat{x}_r, \hat{y}_r, \hat{g}_r, \hat{L}_v) \\ &= \arg \min_{x_r, y_r, g_r, L_v} \sum_{l=1}^{L_v} \left(\tilde{P}_{r,I(l)} - g_r s_{I(l)}\right)^2 + \sum_{l=L_v+1}^L \tilde{P}_{r,I(l)}^2 \\ &= \arg \min_{x_r, y_r, g_r, L_v} \sum_{l=1}^{L_v} \left[\left(\tilde{P}_{r,I(l)} - g_r s_{I(l)}\right)^2 - \tilde{P}_{r,I(l)}^2\right] \end{aligned}$$



$$\begin{aligned}
&= \arg \min_{x_r, y_r, g_r, L_v} \sum_{l=1}^{L_v} \left[ \left( g_r s_{I(l)} \right)^2 - 2 \tilde{P}_{r,I(l)} g_r s_{I(l)} \right] \\
&= \arg \min_{x_r, y_r, g_r, L_v} g_r^2 \|s_I\|^2 - 2 g_r \tilde{P}_{r,I}^T s_I. \quad (17)
\end{aligned}$$

Given  $\hat{x}_r, \hat{y}_r$  and  $\hat{L}_v$  in problem (17), we have

$$\hat{g}_r = \frac{\tilde{P}_{r,I}^T s_I}{\|s_I\|^2}. \quad (18)$$

Using (18), problem (17) can be further rewritten as

$$\begin{aligned}
&(\hat{x}_r, \hat{y}_r, \hat{L}_v) \\
&= \arg \min_{x_r, y_r, L_v} \left( \frac{\tilde{P}_{r,I}^T s_I}{\|s_I\|^2} \right)^2 \cdot \|s_I\|^2 - 2 \frac{\tilde{P}_{r,I}^T s_I}{\|s_I\|^2} \cdot \tilde{P}_{r,I}^T s_I \\
&= \arg \max_{x_r, y_r, L_v} \frac{\tilde{P}_{r,I}^T s_I}{\|s_I\|}. \quad (19)
\end{aligned}$$

Through the above transformations, the optimization problem (11) with respect to  $2L + 3$  variables is equivalently simplified as the optimization problem (19) with respect to 3 variables, which greatly reduces the computational complexity. We can see that the optimization objective of the proposed VLP scheme in (19) is to maximize the angle between the frequency-domain valid RSS modulus vector  $\tilde{P}_{r,I}$  and the valid RSS characteristic vector  $s_I$ . In other words, the direction of the valid RSS characteristic vector, i.e.,  $\frac{s_I}{\|s_I\|}$ , is adopted as the key feature for estimating the receiver horizontal coordinate  $(x_r, y_r)$ . Besides, the proposed scheme only requires the ratios of  $\{g_l\}_{l=1}^L$  and values of  $\{M_l\}_{l=1}^L$  for positioning.

In this work, we consider the benchmark scheme which assumes that the number of valid LEDs is fixed (i.e.,  $L_v = L_{v0}$ ) [26]. By jointly estimating the receiver position and receiver coefficient, the benchmark scheme is more robust to the transmission gain compared with the conventional scheme such as least squares (LS) scheme. The corresponding estimation problem is expressed as

$$\begin{aligned}
(\hat{x}_r, \hat{y}_r) &= \arg \min_{x_r, y_r, g_r} \|\tilde{P}_{r,L_{v0}} - g_r s_{L_{v0}}\|^2 \\
&= \arg \max_{x_r, y_r} \frac{\tilde{P}_{r,L_{v0}}^T s_{L_{v0}}}{\|s_{L_{v0}}\|}, \quad (20)
\end{aligned}$$

where  $\tilde{P}_{r,L_{v0}}$  and  $s_{L_{v0}}$  represent the frequency-domain valid RSS modulus vector and valid RSS characteristic vector assuming  $L_{v0}$  valid LEDs. It is obvious that the common assumption of sufficiently large FOV corresponds to the case of  $L_{v0} = L$ .

Note that problems (19) and (20) are non-convex, and it is difficult to obtain their optimal solutions. Here, we solve problems (19) and (20) via brute-force search (BFS) in the 2D receiver plane. Take problem (19) for example. Let  $F(x_r, y_r, L_v)$  denote the objective function in the optimization problem. We divide the receiving plane into multiple discrete grid points  $\{(x_r^{(j)}, y_r^{(j)})\}_{j=1}^J$ , where  $J$  denotes the number of grid points. Then, we find the corresponding  $\hat{L}_v^{(j)} \in \{3, \dots, L\}$  with the largest  $F(x_r^{(j)}, y_r^{(j)}, L_v^{(j)})$  for each grid

### Algorithm 1 The Proposed VLP Scheme

- 1: Input the LED coordinates  $\{c_l\}_{l=1}^L$ , attenuation exponents  $\{M_l\}_{l=1}^L$ , ratios of transmitter characteristic scalars  $\{g_l\}_{l=1}^L$ , and discrete-time signal samples  $\{p_{r,l}[n]\}_{l=1,n=1}^{L,N}$ , discrete grid points  $\{(x_r^{(j)}, y_r^{(j)})\}_{j=1}^J$ .
- 2: Perform  $N$ -point DFT on  $\{p_{r,l}[n]\}_{l=1,n=1}^{L,N}$  to obtain  $\{P_{r,l}\}_{l=1}^L$ . Let  $\tilde{P}_{r,l} = |P_{r,l}|$ .
- 3: **for** Each grid point  $(x_r^{(j)}, y_r^{(j)})$  **do**
- 4: Find the corresponding  $\hat{L}_v^{(j)} \in \{3, \dots, L\}$  with the largest  $F(x_r^{(j)}, y_r^{(j)}, L_v^{(j)})$ .
- 5: **end for**
- 6: Find the grid point  $(\hat{x}_r, \hat{y}_r)$  with the largest  $F(x_r^{(j)}, y_r^{(j)}, \hat{L}_v^{(j)})$ . Output the estimated receiver coordinate  $\hat{c}_r = [\hat{x}_r, \hat{y}_r, h_r]^T$ .

TABLE I  
THE EXPERIMENTAL PARAMETERS FOR INDOOR VLP SYSTEM

| Parameters                      | Value                                          |
|---------------------------------|------------------------------------------------|
| Number of LEDs                  | 5                                              |
| Coordinates of LEDs             | $(\pm 0.3, \pm 0.3, 2.5)$ m<br>$(0, 0, 2.5)$ m |
| Number of PD-EST positions      | 9                                              |
| Number of PD-POS positions      | 49                                             |
| DC voltage of LEDs              | 15 V                                           |
| AC frequency of LEDs            | 500 kHz                                        |
| AC peak-to-peak voltage of LEDs | 10 V                                           |
| Height of APD                   | 1.0 m                                          |
| DC voltage of APD               | 5 V                                            |

point  $(x_r^{(j)}, y_r^{(j)})$ . Finally, the grid point  $(\hat{x}_r, \hat{y}_r)$  with the largest  $F(x_r^{(j)}, y_r^{(j)}, \hat{L}_v^{(j)})$  is selected as the estimated receiving position. As for problem (20), we simply set  $\hat{L}_v^{(j)} = L$ , and other steps are the same as those for solving (19). The overall procedure of the proposed VLP scheme is summarized in Algorithm 1.

Let  $M$  denote the number of discrete grid points. The computational complexity of the proposed scheme with BFS is  $\mathcal{O}(M \cdot L^2)$ . The computational complexity of the benchmark scheme with BFS is  $\mathcal{O}(M \cdot L)$ .

## IV. EXPERIMENTAL RESULTS AND DISCUSSIONS

In this section, experimental results are presented to evaluate the performance of the proposed VLP scheme under limited receiving FOV.

### A. Experimental Setup

Figure 3 illustrates the experimental environment in an enclosed room of dimension 9.3 m  $\times$  3.7 m  $\times$  2.7 m (length  $\times$  width  $\times$  height). In the experiments, we adopt 5 LEDs (Kecent SL-COB, labeled as LED “A”  $\sim$  LED “E”) with coordinates  $(\pm 0.3, \pm 0.3, 2.5)$  m and  $(0, 0, 2.5)$  m. All LEDs are driven by the same DC bias and sinusoid signal using a bias-tee circuit. The frequency of sinusoid signal is set to be 500 kHz for good separation with the near-DC component

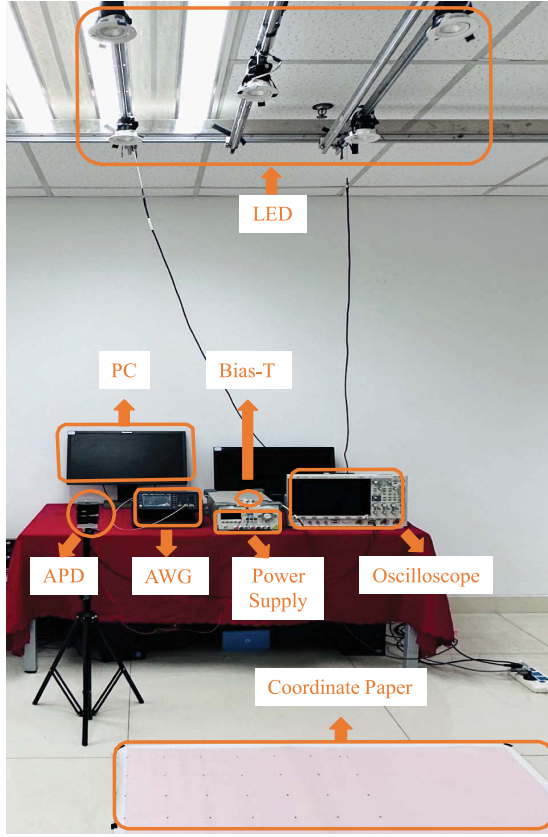


Fig. 3. The experimental environment.

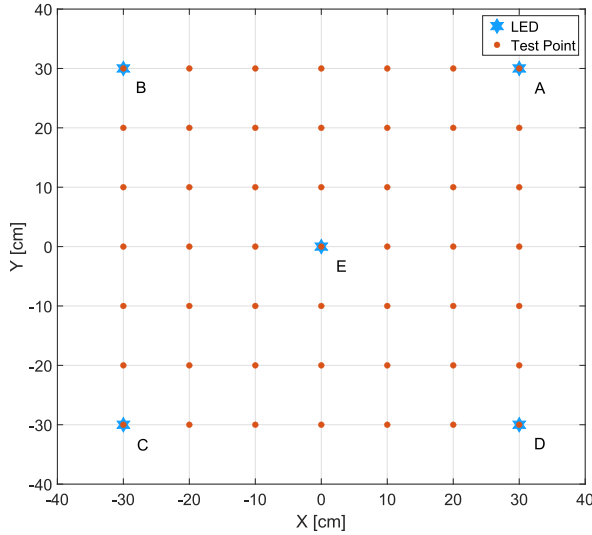


Fig. 4. The horizontal positions of LEDs and test receiving points.

including DC bias and ambient light. At the receiver, an AC-coupled avalanche photodiode (APD, C12702-11) is placed at the height  $h_r = 1$  m to receive the optical signals and output the electrical AC signals. A real-time oscilloscope (Agilent Technologies MSO-X 6004A) is used to collect the output signals of APD for further offline processing on computer. The positioning process is similar as in [26]. We first estimate the LED parameters (i.e., ratios of  $\{g_l\}_{l=1}^L$  and values of  $\{M_l\}_{l=1}^L$ ) using one APD (labeled as “PD-EST”). Then, we

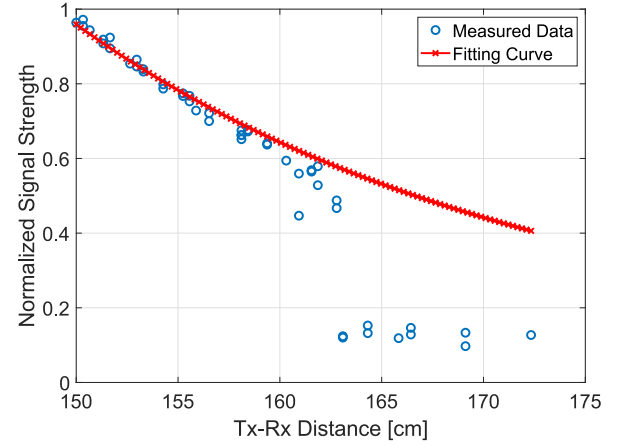


Fig. 5. Comparison of the frequency-domain normalized RSS and the RSS fitting curve of the benchmark scheme.

locate the position of another test APD (labeled as “PD-POS”) based on the estimated LED parameters. The output AC signal of PD-POS are measured at 49 receiving points. Figure 4 depicts the horizontal positions of LEDs and test receiving points. As shown in Fig. 4, the test region is in the range  $[-30, 30]$  cm  $\times$   $[-30, 30]$  cm, i.e., a region of 60 cm  $\times$  60 cm.

Table I lists the main experimental parameters. Unless otherwise specified, the number of signal samples at the receiver is  $N = 4000$ . For comparison, we also present the results of benchmark scheme which assumes three different values of the number of valid LEDs, namely  $L_{v0} = 3$ ,  $L_{v0} = 4$  and  $L_{v0} = 5$ .

*Remark 3:* Standard PIN PD and APD are two common types of PDs. Compared with standard PIN PD, APD has higher sensitivity, gain and larger detection dynamic range. Therefore, the APD is adopted in the experiment.

### B. Experimental Results

Figure 5 compares the normalized frequency-domain RSS in the experiment and the RSS fitting curve given by (9) with  $t_{r,l} = 1, \forall l$ . It can be seen that when the transceiver distance is less than 163 cm (i.e., the horizontal transceiver distance is larger than 63 cm), the measured data can be well fitted by the RSS fitting curve. However, when the transceiver distance exceeds 163 cm, the actual RSS value is close to zero and much smaller than the fitted value. This is because the APD cannot receive the signals from some LEDs when the incidence angles are larger than the receiving FOV. The results imply the inapplicability of assuming all LEDs are valid in real VLP scenarios under limited receiving FOV. Besides, using the results in Fig. 5, the FOV of the adopted APD is measured to be about  $\Phi = 23^\circ$ , and the SNR is estimated to be about 10 dB.

Let  $I$  denote the number of pre-known positions of PD-EST used for LED parameter estimation. Table II lists the estimated values of LED parameters using PD-EST under different  $I$ . Figure 6 shows the mean positioning errors of different VLP schemes. The LED parameters are set to be the estimated values using PD-EST or empirical values. The empirical values of LED parameters are  $g_1 = \dots = g_5 = 1$  and  $M_1 =$

TABLE II  
ESTIMATED VALUES OF LED PARAMETERS USING PD-EST

| $I$ | Ratios of $\{\hat{g}_l\}_{l=1}^L$      | $\{\hat{M}_l\}_{l=1}^L$                |
|-----|----------------------------------------|----------------------------------------|
| 5   | 5.8641, 5.6933, 6.0316, 5.3483, 5.3095 | 1.0000, 1.0752, 1.0287, 1.0229, 0.9901 |
| 7   | 5.6627, 5.7871, 6.1159, 5.6091, 5.4099 | 1.0000, 1.1019, 1.0506, 1.0526, 1.0178 |
| 9   | 5.6983, 5.8278, 6.1944, 5.6560, 5.4530 | 1.0000, 1.1042, 1.0591, 1.0522, 1.0200 |

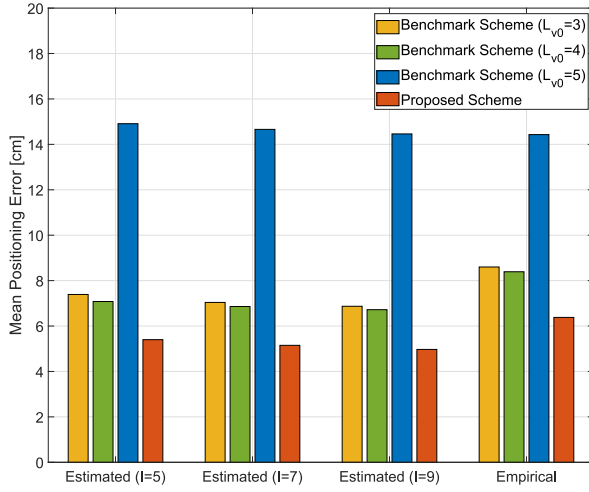


Fig. 6. Mean positioning error using estimated values or empirical values of LED parameters.

TABLE III  
ESTIMATED INVALID LEDs AT TEST RECEIVER POINTS USING THE PROPOSED VLP SCHEME

| $X[\text{cm}] \backslash Y[\text{cm}]$ | -30  | -20 | -10 | 0    | 10 | 20 | 30   |
|----------------------------------------|------|-----|-----|------|----|----|------|
| -30                                    | A    | A   | A   | A, B | B  | B  | B    |
| -20                                    | A    | A   | A   | /    | B  | B  | B    |
| -10                                    | A    | A   | /   | /    | /  | B  | B    |
| 0                                      | A, D | /   | /   | /    | /  | /  | B, C |
| 10                                     | D    | D   | /   | /    | /  | C  | C    |
| 20                                     | D    | D   | D   | /    | C  | C  | C    |
| 30                                     | D    | D   | D   | C, D | C  | C  | C    |

$\dots = M_5 = 5$ . It can be seen from Fig. 6 that the mean positioning errors of the benchmark and proposed schemes decrease slightly as  $I$  increases. To obtain a higher positioning accuracy, we adopt the estimated LED parameters employing PD-EST with  $I = 9$  by default. Besides, the mean positioning error of the proposed scheme is lower than 7 cm using either the estimated values or empirical values of  $\{\hat{g}_l\}_{l=1}^5$  and  $\{\hat{M}_l\}_{l=1}^5$ . The results imply that the proposed VLP scheme can maintain a stable positioning accuracy even under empirical parameter settings.

Noting  $t_{r,l} = \text{rect}(\frac{\varphi_{r,l}}{\Phi})$ , the receiver position and invalid LEDs at each receiving point can be simultaneously estimated using the proposed VLP scheme (19). Specifically, we can estimate the incidence angles  $\{\varphi_{r,l}\}_{l=1}^L$  using the estimated receiver position  $(\hat{x}_r, \hat{y}_r, \hat{h}_r)$  and LED positions  $\{(x_l, y_l, h_l)\}_{l=1}^L$ . Then, the LEDs corresponding to the  $\hat{L}_v$  smallest incidence angles are valid LEDs, while the remaining LEDs are invalid LEDs. Besides, using the measured APD

TABLE IV  
MEAN POSITIONING ERRORS IN THE SIMULATIONS AND EXPERIMENTS UNDER EXPERIMENTAL PARAMETERS

| Mean positioning errors [cm]      | Simulation | Experiment |
|-----------------------------------|------------|------------|
| Benchmark Scheme ( $L_{v0} = 3$ ) | 4.64       | 6.80       |
| Benchmark Scheme ( $L_{v0} = 4$ ) | 3.89       | 6.67       |
| Benchmark Scheme ( $L_{v0} = 5$ ) | 8.60       | 14.36      |
| Proposed Scheme                   | 3.11       | 4.97       |

FOV  $\Phi = 23^\circ$  from Fig. 5, we can obtain the true value of  $t_{r,l}$  for each test point as

$$t_{r,l} = \begin{cases} 1, & \text{if } h/d_{r,l} > \cos \Phi \\ 0, & \text{if } h/d_{r,l} \leq \cos \Phi \end{cases} \quad (21)$$

Table III shows the estimated invalid LEDs at test receiver points using the proposed VLP scheme, where “A”, “B”, “C” and “D” represent the LEDs illustrated in Fig. 4, and “/” represents that there is no invalid LED. It is seen that the estimated invalid LEDs using the proposed VLP scheme are consistent with the true results based on the measured APD FOV. It implies that the proposed VLP scheme yields an accurate LED validity determination at test receiver points, which leads to the robustness of the proposed VLP scheme under limited receiving FOV. The estimated positions of 49 test points are shown in Fig. 7. It can be seen that the estimated positions of the benchmark and proposed schemes are close when  $L_{v0}$  matches the estimated value  $\hat{L}_v$ . However, the estimated position of the benchmark scheme with fixed  $L_{v0}$  can deviate greatly from the true position in the existence of invalid LEDs. The large position deviation using the benchmark scheme mainly results from two aspects: (1) With larger  $L_{v0}$ , some invalid LEDs are adopted for positioning, introducing ineffective RSS information; (2) With smaller  $L_{v0}$ , some valid LEDs are discarded in the positioning, losing some effective RSS information. The results reveal the necessity of LED validity determination for RSS-based VLP under limited receiving FOV.

Figure 8(a) illustrates the cumulative distribution function (CDF) of the experimental positioning errors using the benchmark and proposed schemes. It can be seen that the positioning error of the benchmark scheme is larger than that of the proposed scheme. Specifically, the maximum positioning errors of the benchmark scheme with  $L_{v0} = 3/4/5$  and the proposed scheme are about 14 cm/18 cm/27 cm and 12 cm, while their mean positioning errors are 6.87 cm/6.72 cm/14.36 cm and 4.97 cm. The results show the performance degradation of the benchmark scheme under limited receiving FOV due to the use of invalid LEDs or discard of valid LEDs. In contrast, the proposed scheme can effectively improve the positioning accuracy by jointly

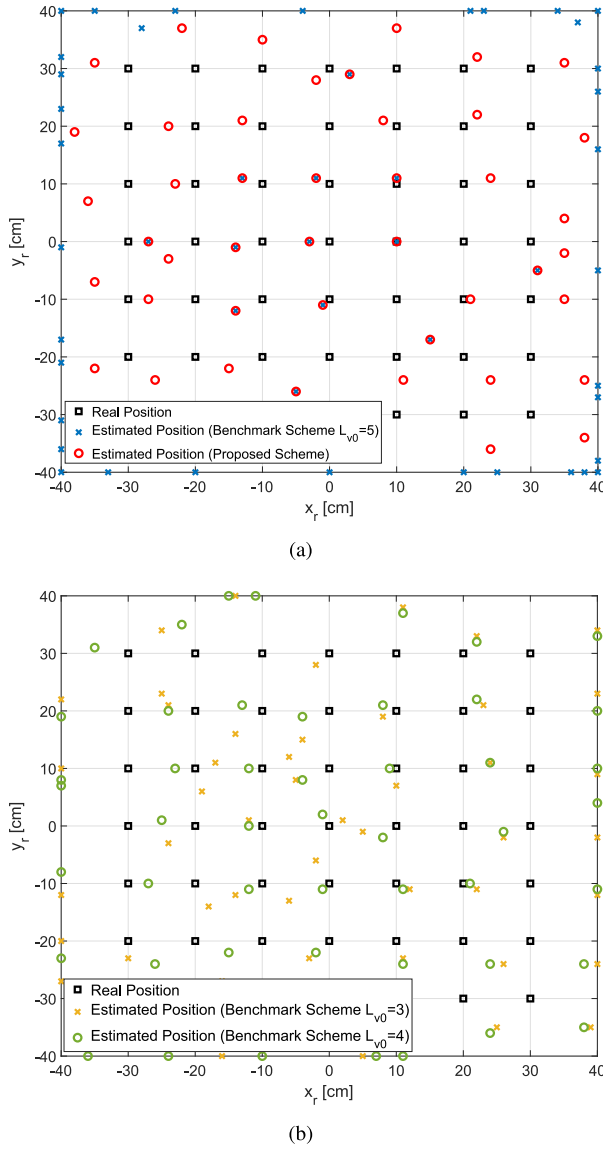


Fig. 7. Estimated positions of 49 test points. (a) Using the proposed scheme and the benchmark scheme with  $L_{v0} = 5$ . (b) Using the benchmark scheme with  $L_{v0} = 3$  and  $L_{v0} = 4$ .

estimating  $(x_r, y_r)$  and  $L_v$ . Figure 8(b) illustrates the CDF of the simulated positioning errors using the same parameters in the experiments. Table IV lists the mean positioning errors. Despite the fact that the experimental positioning errors are a little larger than the simulated positioning errors due to the measurement error and parameter estimation error in the experiments, the simulated positioning errors have a similar distribution as the experimental positioning errors. It implies that the simulation results can provide guidance to the evaluation of positioning accuracy. The results in Fig. 7, Fig. 8 and Table IV verify that the proposed VLP scheme can maintain a high positioning accuracy under limited receiving FOV, which is suitable for practical applications.

To investigate the effect of the number  $N$  of signal samples on the positioning accuracy, we collect the received signal in one period with  $N \in \{40, 400, 1000, 2000, 4000\}$ . Figure 9 illustrates the mean positioning error versus  $N$ . Obviously, the mean positioning errors of the benchmark and

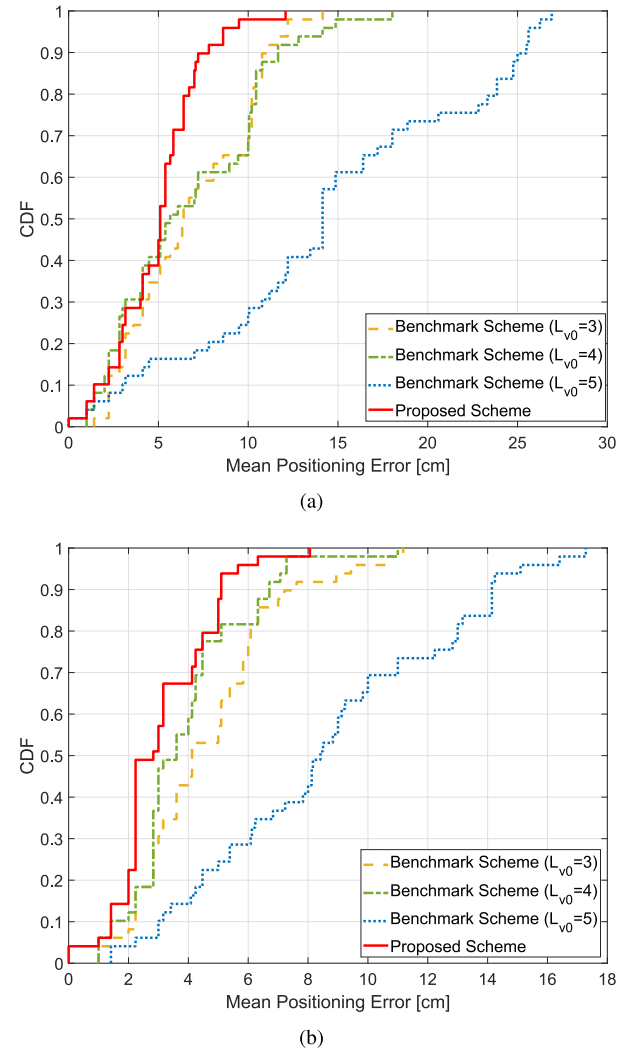


Fig. 8. CDF of positioning errors with different schemes. (a) Experimental results. (b) Simulation results.

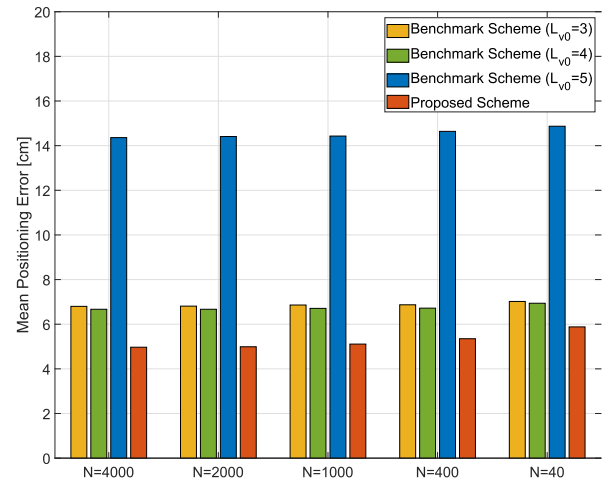


Fig. 9. Mean positioning error versus the number  $N$  of signal samples.

proposed schemes increase with  $N$ . Although increasing  $N$  can improve the positioning accuracy, it can also increase the computational complexity. Therefore, it is essential to select



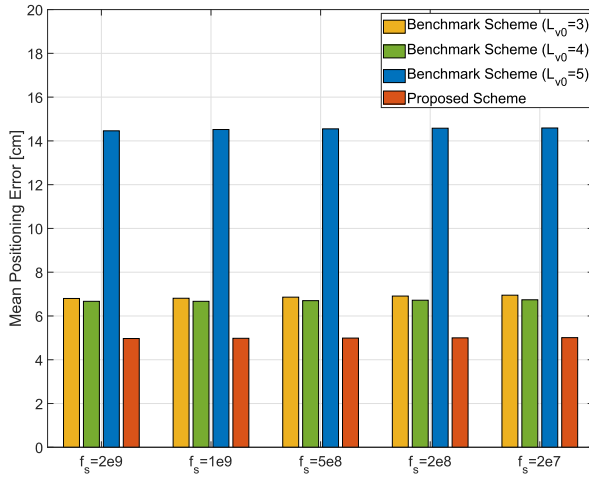


Fig. 10. Mean positioning error versus the sampling rate  $f_s$  with  $N = 4000$ .

an appropriate  $N$  in practical applications to achieve a good complexity-performance tradeoff.

Figure 10 depicts the mean positioning error versus the sampling rate  $f_s$  with  $N = 4000$ . It can be seen that the proposed VLP scheme achieves a stable positioning accuracy under different sampling rates with fixed  $N$ . Comparing the results of Figs. 9 and 10, we can see that the number  $N$  of signal samples has a larger impact on the positioning accuracy than the sampling signal rate  $f_s$ . It implies that we can select appropriate  $N$  for positioning without the requirement of high sampling rate in real scenarios.

## V. SIMULATION RESULTS AND DISCUSSIONS

As mentioned above, the similar distributions of simulated and experimental positioning errors imply that the simulation results can provide guidance to the evaluation of positioning accuracy. Thus, we present more simulation results in this section to further evaluate the performance of the proposed VLP scheme.

### A. Simulation Setup

In the simulations, we consider the scenario where  $L = 9$  LEDs are installed on the ceiling of a room. The coordinates of the nine LEDs are  $(\pm 1, \pm 1, 2)$  m,  $(0, \pm 1, 2)$  m,  $(\pm 1, 0, 2)$  m and  $(0, 0, 2)$  m. The LED parameters are set to be  $g_1 = \dots = g_5 = 1$  and  $M_1 = \dots = M_5 = 5$ . The PD is located in the  $2 \text{ m} \times 2 \text{ m}$  horizontal plane with height  $h_r = 0$  m. The range of test receiving points is  $[0, 1] \times [0, 1] \text{ m}^2$ . The AC frequency of the transmitted signal is  $f = 500$  kHz. The detector area of the PD is  $A_r = 7.07 \text{ mm}^2$  and detection gain is  $R = 1$ . The FOV of PD is  $\Phi = 30^\circ$ . Unless otherwise specified, the signal-to-noise ratio (SNR) is set to be  $10 \lg \frac{B A_r g_l}{2\pi h^2 z_{r,l}} = 20$  dB, and the number of signal samples is set to be  $N = 4000$ . The sampling rate is  $f_s = 200$  MHz. The signal phase of each LED is uniformly chosen from  $[0^\circ, 360^\circ)$ . Due to the symmetry of LED deployment, we only consider the receiving region in the first quadrant. Specifically, we investigate the positioning accuracy for the receiving points in the following four cases:

- 1) Case 1: Receiving points on the X-axis.
- 2) Case 2: Receiving points on the line  $y = 1$  m.

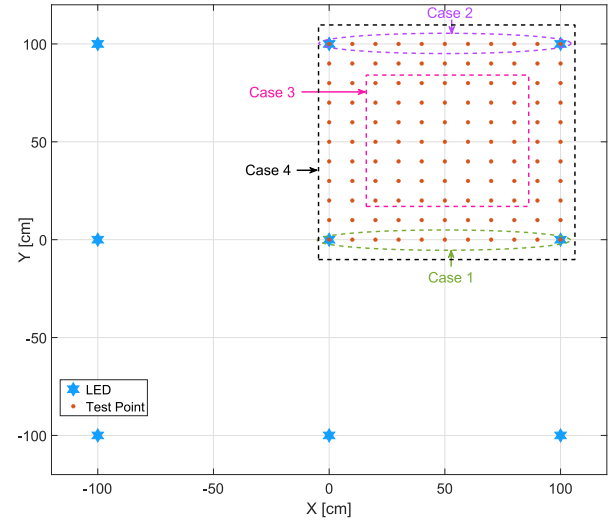


Fig. 11. The horizontal positions of LEDs and test receiver points.

- 3) Case 3: Receiving points in the square region of  $[0.2, 0.8] \times [0.2, 0.8] \text{ m}^2$ .
- 4) Case 4: Receiving points in the square region of  $[0, 1] \times [0, 1] \text{ m}^2$ .

Figure 11 illustrates the horizontal positions of the LEDs and test receiving points.

### B. Simulation Results

Figure 12 shows the simulated positioning errors of the benchmark and proposed schemes versus the X-coordinates of receiving points in Cases 1 and 2. It can be seen that the positioning error of the benchmark scheme varies greatly with the values of  $L_{v0}$ . The mean positioning error of the benchmark scheme with  $L_{v0} = 4$  is smaller than those with  $L_{v0} = 3$  and  $L_{v0} = 5$  in Case 1, while the mean positioning error of the benchmark scheme with  $L_{v0} = 3$  is smaller than those with  $L_{v0} = 4$  and  $L_{v0} = 5$  in Case 2. Besides, the benchmark scheme can show a sudden increase of the positioning error in Fig. 12, as some invalid LEDs are adopted for positioning, introducing ineffective RSS information. Take the results of the benchmark scheme with  $L_{v0} = 5$  in Fig. 12(a) as an example. At receiver point  $(30, 0, 0)$ , there are 4 valid LEDs and 1 invalid LED. The PD cannot receive the transmitted signal from the invalid LED, and its output is merely the noise. When adopting the invalid LED for positioning, the low-amplitude PD noise is mistakenly considered as useful signal with a long transmission distance, yielding an estimated receiver position far from the invalid LED. In contrast, the positioning error of the proposed scheme is more stable due to the LED validity determination. Specifically, the proposed scheme achieves a stable positioning accuracy below 3 cm for each receiving point in Cases 1 and 2. The results imply that the benchmark scheme with fixed  $L_{v0}$  is not suitable for real scenarios under limited receiving FOV.

Note that for the receiving points in Case 3, the real number of valid LEDs is constant at 4. We simulate the positioning error of the benchmark scheme with  $L_{v0} \in [4, 9]$  in Case 3. Figure 13 shows the mean positioning errors of the benchmark and proposed schemes versus the values of

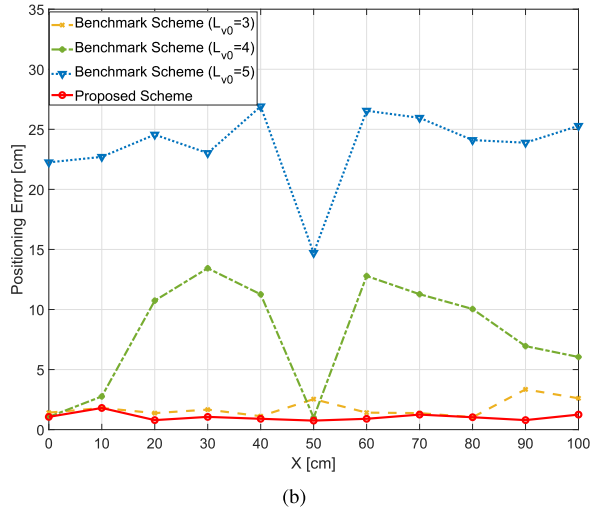
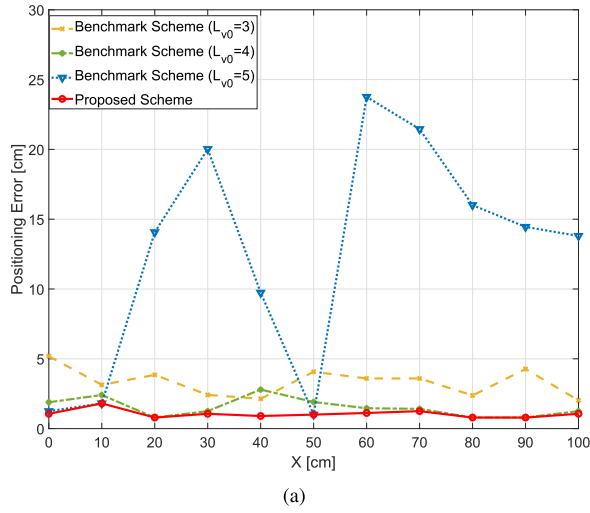


Fig. 12. Simulated positioning error versus the X-coordinate of receiving point. (a) Case 1. (b) Case 2.

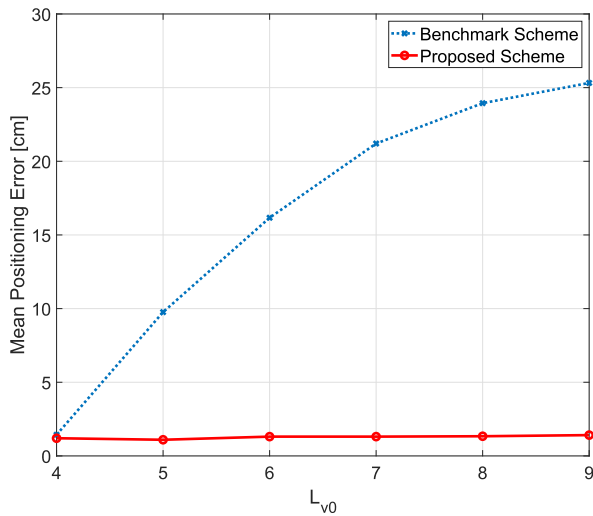


Fig. 13. Mean positioning error versus  $L_{v0}$  in Case 3.

$L_{v0}$  in Case 3. It can be seen that the mean positioning error of the benchmark scheme increases rapidly with  $L_{v0}$  due to the use of invalid LEDs. In contrast, the mean positioning

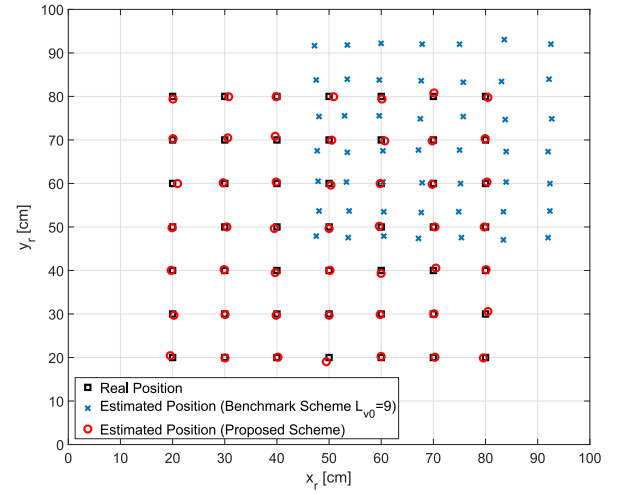


Fig. 14. Estimated positions of 49 test points in Case 3.

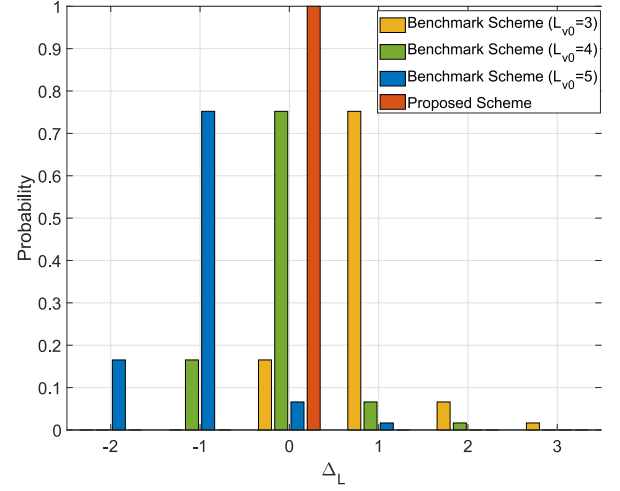


Fig. 15. Probability distribution of  $\Delta_L$  using different schemes in Case 4.

error of the proposed scheme is not affected by the number of invalid LEDs, yielding a stable positioning accuracy. The results imply that the proposed VLP scheme is more robust than the benchmark scheme under limited receiving FOV.

Figure 14 shows the estimated positions of 49 test points in Case 3. The benchmark scheme assumes  $L_v = L_{v0} = 9$ . It can be seen that the estimated position using the proposed VLP scheme is close to the true position, while the estimated position using the benchmark scheme deviates greatly from the true position towards the direction away from the invalid LEDs. This is because that the benchmark scheme mistakenly adopts noises as useful signals from invalid LEDs for positioning, which yields an estimated position far from the invalid LEDs based on the low-amplitude noises.

Denote the difference between the true value of  $L_v$  and the estimated value  $\hat{L}_v$  by  $\Delta_L$ . Figure 15 shows the probability distribution of estimation error  $\Delta_L$  using the benchmark and proposed schemes in Case 4. It is obvious that estimated value  $\hat{L}_v$  using the proposed scheme matches the true value of  $L_v$  perfectly, while the benchmark scheme yields a high probability of incorrectly estimating  $L_v$ . Specifically, the probability

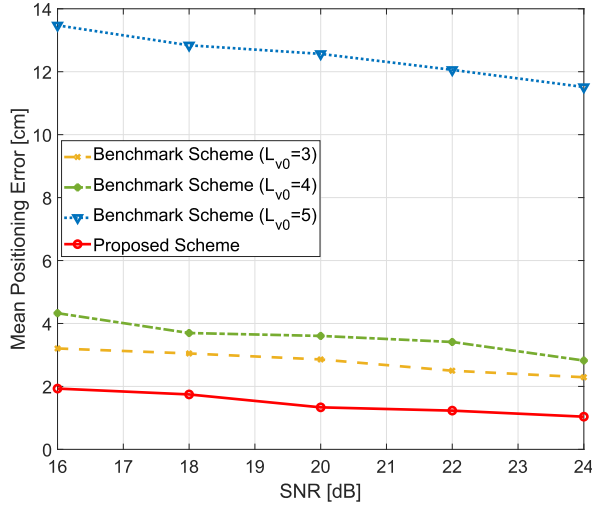


Fig. 16. Mean positioning error versus SNR in Case 4.

of non-zero  $\Delta_L$  using the benchmark scheme exceeds 0.2 for  $L_{v0} = 3, 4, 5$ . The results verify the effectiveness of the proposed scheme in LED validity determination. Figure 16 shows the mean positioning error versus SNR in Case 4. Obviously, the mean positioning error decreases with SNR. The proposed VLP scheme yields a lower mean positioning error compared with the benchmark scheme, as a result of more accurate LED validity determination.

## VI. CONCLUSION

In this paper, we have proposed a robust RSS-based VLP scheme under limited receiving FOV. The proposed scheme first performs the time-frequency transform on the received time-domain signal, and then jointly estimates the receiver coordinate, LED validity indicator vector, signal phase vector and receiving coefficient from the frequency-domain RSS vector. Through mathematical derivation, the original estimation problem is equivalently transformed into the problem that maximizes the angle between the valid RSS modulus vector and the valid RSS feature vector. Experimental and simulation results have verified the effectiveness of the proposed scheme under limited receiving FOV, which results from the accurate LED validity determination and makes the proposed scheme appealing for practical VLP applications. Future work includes the extension to the multiple-PD case and low-complexity processing algorithm.

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**Feng Chen** received the B.S. and M.S. degrees from the University of Science and Technology of China, Hefei, China, in 2020 and 2023, respectively. Her research interests include visible light positioning and optical network.



**Changyuan Liu** received the B.S. degree in mathematics and applied mathematics from the University of Science and Technology of China, Hefei, China, in 2023, where he is currently pursuing the M.S. degree. His research interests include visible light positioning and visible light communication.



**Yan-Yu Zhang** received the B.S. degree in communication engineering, the M.S. degree in communication and information systems, and the Ph.D. degree in information and communication engineering from National Digital Switching System Engineering and Technological Research Center, Zhengzhou, China, in 2009, 2012, and 2016, respectively, where he is currently a Lecturer. His current research interests are in energy-efficient signal designs for radio frequency and optical wireless communication systems.



**Yi-Jun Zhu** received the B.Eng., M.Sc., and Ph.D. degrees from the National Digital Switching System Engineering and Technological Research Center, Zhengzhou, China, in 1999, 2002, and 2010, respectively. In 2011, he visited the Department of Electrical and Computer Engineering, McMaster University. He is currently with the School of Information Engineering University. His research interests include the areas of wireless communication theory, visible light communications, and signal processing.



**Nuo Huang** received the B.S. degree in electronics and information engineering from the Huazhong University of Science and Technology, Wuhan, China, in 2012, and the Ph.D. degree in information and communication engineering from the National Mobile Communications Research Laboratory, Southeast University, Nanjing, China, in 2019. He is currently an Associate Professor with the Department of Electronic Engineering and Information Science, University of Science and Technology of China, Hefei, China. From December

2015 to June 2017, he was a Visiting Student with the Department of Electrical Engineering, Columbia University, New York, NY, USA. He was selected by the National Postdoctoral Program for Innovative Talents in 2019. His research interests include resource allocation and transceiver design in wireless (optical) communications.



interests include wireless communications, optical wireless communications, and signal processing. He was awarded by Hong Kong Qiushi Outstanding Young Researcher Award in 2016.